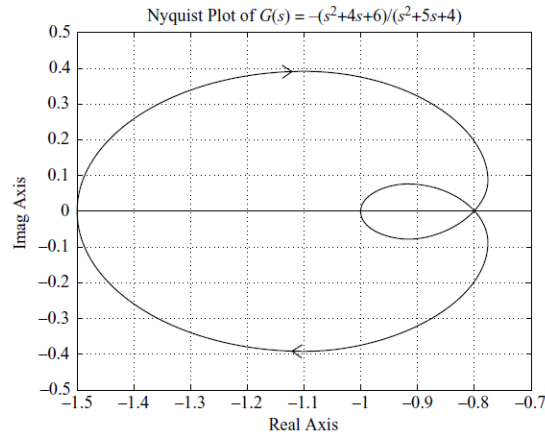


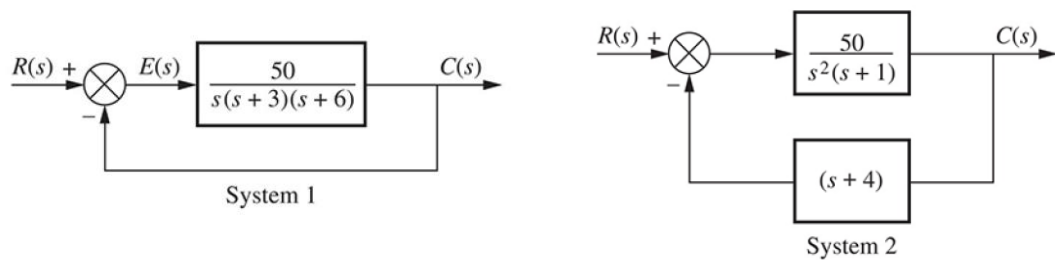
R0004E Exercise 4:

- Determine the stability of the following systems, based on the Nyquist criteria:



- Draw the Nyquist diagram for each of the following system and determine the value of P, N and Z, and the stability of the closed loop system.

Hint: Follow the pattern of the Nyquist plot in the summary plot given as a supplement for Lecture 9.



- For the following system:

$$G(s) = \frac{K}{(s+2)(s+4)(s+6)}$$

- Plot the Nyquist diagram for $K=1$.
 - Find the range of the value of K for which the unity feedback system is stable.
- Consider the open loop system:

$$G(s) = \frac{4}{s(s+2)}$$

Design a lead compensator for the system so that the static velocity error constant K_v is 20 sec^{-1} , the phase margin is at least 50° , and the gain margin is at least 10 dB.